

Temperature-Controlled Soldering Stations

You may be tempted to buy a cheap plug-in soldering iron in most cases but the damage that it might cause would be much more than the price of a temperature-controlled soldering station



ANKIT GUPTA

The cheap unregulated soldering irons do not have any kind of thermal regulation. These directly plug into the wall socket and heat up to a high temperature while sitting in the holder. There is some amount of natural thermal regulation in such soldering irons but that is due to the physical environment; the more the tip gets hotter, the more heat it dissipates, and similarly the more it cools down, the less heat is dissipated. These soldering irons may be described as thermally-balanced but the output is neither controlled nor stable.

Such soldering irons heat up to high temperatures, so the initial soldering occurs at way too high temperature, which leads to poor-quality soldering and possible damage to the compo-

nents. With a temperature-controlled iron, you can set the temperature beyond the melting point of your specific solder but still within a safe limit for the components. Especially, with SMD

packages, using too high temperature or holding the iron for too long in position (because of inefficient heat transfer) may damage them.

When the soldering tip touches the joint, its temperature drops as the heat is dissipated in the joint. If the part you are soldering can dissipate heat at a rate more than what your soldering iron can deliver, the temperature will keep dropping until it can't melt solder any more. This is normally the case when you are soldering components on the power planes with improper thermal relief or thick joints.

Fig. 1 shows the different thermal reliefs. These are normally required with power planes. If there is no thermal relief, the heat is dissipated at a high rate due to the larger surface area. The tip cools down rapidly and the joint is dry soldered. Temperature-controlled irons are useful in such a case as these can dump extra power into the bit without increasing the tempera-

Solder types and their melting temperatures

Solder is a metallic alloy that, when melted with the heat from a soldering iron, can join metallic surfaces together. The most common solder compositions used for electronics work have tin (Sn), lead (Pb) and silver (Ag) elements.

The most widely used composition is 60/40 (Sn/Pb). This composition doesn't melt on a specific temperature. It melts at a range of temperatures as shown in the table.

63/37 is the more recent type and it melts on a specific temperature. The same applies to solidification too.

The third composition (62/36/2) is used for professional-quality products. For regular electronics works, 63/37 is recommended.

These compositions are not used in European countries where lead-free is a primary requirement.

Most Common Solder Compositions

Type	Tin (Sn)	Lead (Pb)	Silver (Ag)	Melting temperature
60/40	60%	40%	0%	183-190°C
63/37	63%	37%	0%	183°C
62/36/2	62%	36%	2%	179°C